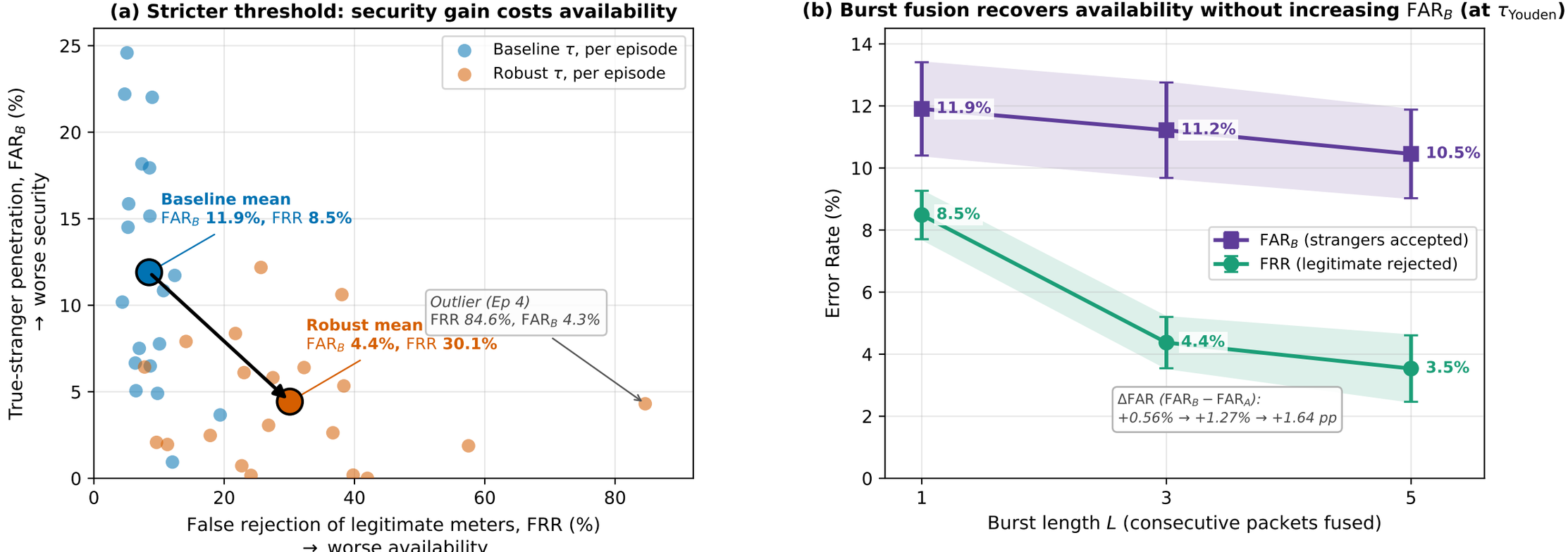


Fig. 1. Security–availability trade-off and multi-packet burst recovery under the open-set AMR gateway protocol ($N = 26{,}000$ raw transmissions across 20 seeded evaluation episodes). Panel (a): Large markers denote episode means; tightening the decision boundary to τ_{robust} reduces stranger penetration (FAR_B : 11.9% → 4.4%) at the cost of a severe legitimate false rejection surge (FRR : 8.5% → 30.1%). Panel (b): Under baseline calibration (τ_{Youden}), multi-packet burst fusion cuts legitimate rejection from 8.5% ($L = 1$) down to 3.5% ($L = 5$) without increasing stranger acceptance (FAR_B : 11.9% → 10.5%). Shaded bands and error bars indicate ±1 standard error over 20 episodes.